

SCSE3040 Machine Learning Operations – Practical 05

Submission

1. Identification

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Batch 89
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2. Repository link

<https://github.com/shauryaa/aiml-lab-assignments>

3. config.yaml

```
# Settings for the delivery-time model.  
# Change values here, never in the code.  
  
data:  
  path: ../data/delivery_times.csv  
  features:  
    - distance_km  
    - prep_time_min  
    - traffic_level  
    - rain  
  target: delivery_min  
  
split:  
  test_size: 0.2  
  seed: 42  
  
model:  
  name: linear_regression
```

4. Terminal screenshot – pytest run

```
Windows PowerShell
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\MLOps\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\MLOps\P05-config-tests\work
plugins: anyio-4.14.2
collected 10 items

test_model.py::test_rain_never_makes_delivery_faster PASSED [ 10%]
test_model.py::test_further_is_never_faster PASSED [ 20%]
test_model.py::test_error_is_better_than_guessing PASSED [ 30%]
test_model.py::test_heavier_traffic_is_never_faster PASSED [ 40%]
test_orders.py::test_minutes_per_km_simple PASSED [ 50%]
test_orders.py::test_minutes_per_km_rejects_zero_distance PASSED [ 60%]
test_orders.py::test_good_order_is_accepted PASSED [ 70%]
test_orders.py::test_bad_traffic_level_is_rejected PASSED [ 80%]
test_speed.py::test_slow_delivery PASSED [ 90%]
test_speed.py::test_negative_distance_raises PASSED [100%]

===== 10 passed in 0.15s =====
PS C:\MLOps\P05-config-tests\work>
```

5. MAE before and after the refactor

Before (hardcoded settings): MAE 1.92 minutes → After (config-driven): MAE 1.92 minutes – identical, since the refactor only moved the same test_size and seed values into config.yaml without changing them.